

②

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

$$C = \frac{4\pi\epsilon_0 a(2a)}{a}$$

$$2 \cdot 10^{-6} = 4\pi\epsilon_0 \cdot 2 \cdot a$$

$$a = 8991,80 \text{ [m]}$$

$$\Rightarrow b = 17983,61 \text{ [m]}$$

$$V = \frac{4}{3}\pi(b^3 - a^3)$$
$$= 2,13 \cdot 10^{13} \text{ m}^3$$